

Short-Circuit Withstand of Insulated Cables: A Practical Engineering Primer

What the rating means, where the equations come from, how to apply them, and the mistakes that quietly undersize — or oversize — a cable.

Why short-circuit withstand matters

A cable is sized first for the load it carries continuously. But the moment that governs whether the cable survives its lifetime is not the steady load — it is the fault. During a short circuit, current many times the rated value floods the conductor for a fraction of a second before protection clears it. In that brief window the conductor heats almost instantaneously, and if it gets hot enough it damages the insulation against it: charring, voids, loss of dielectric strength, eventual premature failure.

Short-circuit withstand is the cable's ability to survive that thermal pulse without exceeding the temperature its insulation can tolerate. Getting it right is a safety and reliability requirement, not an optimization. Two questions fall out of it:

- **Verification:** can an installed cable survive the fault current available at its location, for as long as the protection takes to clear it?
- **Selection:** what conductor size is needed to survive a given fault?

Both are answered by the same small set of equations. This primer develops them from first principles, shows how to apply them, and flags the assumptions that matter.

The adiabatic principle

The foundational simplification is that a fault is *fast*. Cleared in a few cycles up to a second or two, there is not enough time for meaningful heat to escape the conductor into the surrounding insulation. So all of the I^2R heat generated during the fault is assumed to stay in the metal, raising its temperature. This is the **adiabatic assumption** — no heat loss during the fault.

It is the right assumption for short-circuit work because it is both simple and conservative: by ignoring the heat that genuinely does leak away, it slightly *underestimates* the cable's true capability, erring safe. (For long-duration faults the assumption becomes overly conservative — more on that later.)

From an energy balance — heat generated equals heat stored in the metal's mass — and integrating the temperature-dependent resistance of the metal over the temperature rise,

one arrives at the standard withstand relationship.

The governing equation: ICEA P-32-382

For insulated conductors, the adiabatic withstand is expressed by ICEA P-32-382:

$$\left(\frac{I}{A}\right)^2 t = K \log_{10} \left(\frac{T_2 + \lambda}{T_1 + \lambda} \right)$$

Symbol	Meaning	Units
I	RMS symmetrical short-circuit current	A
A	conductor cross-section	circular mils (cmil)
t	fault duration	s
T ₁	initial (continuous operating) temperature	°C
T ₂	maximum allowable (short-circuit) temperature	°C
K, λ	material constants	—

The constants embed each metal's specific heat, density, and resistivity, and the offset λ is the inferred temperature of zero resistance:

- **Copper:** K = 0.0297, λ = 234
- **Aluminum:** K = 0.0125, λ = 228

A note on units: ICEA works in **circular mils**. One thousand circular mils is 1 kcmil; to convert from metric, multiply mm² by 1973.5 to get cmil. Standard AWG areas (e.g. 4/0 = 211,600 cmil) are tabulated and worth keeping handy.

The three rearrangements you actually use

The equation answers three different practical questions depending on what you solve for:

Allowable current for a known conductor and clearing time:

$$I = A \sqrt{\frac{K \log_{10} \left(\frac{T_2 + \lambda}{T_1 + \lambda} \right)}{t}}$$

Minimum conductor size for a known fault current and clearing time:

$$A = I \sqrt{\frac{t}{K \log_{10} \left(\frac{T_2 + \lambda}{T_1 + \lambda} \right)}}$$

Maximum clearing time a given conductor can tolerate at a given current:

$$t_{max} = \left(\frac{A}{I} \right)^2 K \log_{10} \left(\frac{T_2 + \lambda}{T_1 + \lambda} \right)$$

The short-circuit factor: a faster way

Notice that everything except I, A, and t is fixed once you choose the metal and the two temperatures. Manufacturers exploit this by pre-computing that block into a single **short-circuit factor**, F_C , reducing the whole equation to:

$$\boxed{I_{SC} = \frac{A \cdot F_C}{\sqrt{t}}}, \quad F_C = \sqrt{K \log_{10} \left(\frac{T_2 + \lambda}{T_1 + \lambda} \right)}$$

Representative factors for common metal/insulation/temperature combinations:

Insulation ($T_1 \rightarrow T_2$)	Copper F_C	Aluminum F_C
Thermoset XLPE/EPR (90 $\rightarrow$ 250 °C)	0.0719	0.0470
Thermoset XLPE/EPR (105 $\rightarrow$ 250 °C)	0.0678	0.0443
Thermoplastic PVC/PE (75 $\rightarrow$ 150 °C)	0.0529	0.0346

For these factors to be valid, the conductor must be assumed to start at its full continuous temperature T_1 when the fault strikes — the worst case, and the conservative one. If the cable actually runs cooler in service, its true withstand is higher; using T_1 is correct for a worst-case rating.

Temperature limits come from the insulation, not the conductor

T_1 and T_2 are properties of the **insulation system**, which is what the conductor is protecting. The conductor itself could survive far hotter; the limit exists to keep the insulation intact.

Insulation	T_1 continuous (°C)	T_2 short-circuit (°C)
XLPE / TR-XLPE	90	250
EPR	90	250
PVC ($\leq \sim 300 \text{ mm}^2$)	75	160
PVC (larger)	75	140
Paper / PILC	80-85	200-250

Thermoset insulations tolerate the highest short-circuit temperature (250 °C), which is why XLPE and EPR dominate medium-voltage cable. Thermoplastics soften at lower temperatures and carry correspondingly lower ceilings.

Worked examples

Allowable current. A 4/0 AWG aluminum, TR-XLPE cable (211,600 cmil; 90 → 250 °C; $F_C = 0.047$), with protection clearing in 30 cycles (0.5 s):

$$I_{SC} = \frac{211,600 \times 0.047}{\sqrt{0.5}} \approx 14,100 \text{ A} = 14.1 \text{ kA}.$$

Minimum size. A copper, thermoset feeder must survive 30 kA for 0.5 s:

$$A = \frac{30,000 \times \sqrt{0.5}}{0.0719} \approx 295,000 \text{ cmil} \Rightarrow 300 \text{ kcmil}.$$

A 500 kcmil copper thermoset cable at 0.5 s withstands

$$\frac{500,000 \times 0.0719}{\sqrt{0.5}} \approx 50,800 \text{ A} = 50.8 \text{ kA}.$$

The inverse-square-root law on time

The single most important behavioral feature of the withstand equation is its dependence on clearing time: **withstand scales as $1/\sqrt{t}$** . Halve the clearing time and the cable tolerates $\sqrt{2} \approx 1.41$ times more current; quadruple it and the tolerance halves.

For the 4/0 aluminum example:

Clearing time	Withstand
5 cycles (0.083 s)	34.5 kA
10 cycles (0.167 s)	24.4 kA
30 cycles (0.500 s)	14.1 kA
60 cycles (1.00 s)	9.9 kA

This is why a single tabulated "short-circuit current @ 30 cycle" value is only a reference point. The real cable capability is a *curve*, and the design check is performed at the *actual* clearing time of the protection at that location.

The actual design check: withstand vs. available fault

The withstand equation gives only one side of the comparison. A real verification pairs it with the **available fault current** from a system short-circuit study, at the **actual total clearing time** of the upstream protection:

■ **Withstand at $t \geq$ available fault current**, where t is the total clearing time.

Two inputs come from outside the cable datasheet and are easy to get wrong:

The available fault current comes from the system study — source impedance, transformer, the cable's own impedance, and the network behind it. It is not a cable property.

The clearing time is the *total* time to interrupt: relay operating time plus breaker interrupting time, or fuse total-clear time. The 30 cycles printed in datasheets is a tabulation convention, not your project's number. A fast bus might clear in 5 cycles; a time-delayed backup element might take seconds. Because of the $1/\sqrt{t}$ law, using the right t is as important as using the right fault current.

A useful way to document the result is a **cable damage curve**: plot the withstand-vs-time relationship on the same time-current axes as the protective device's curve. The cable is

protected wherever its damage curve sits above the device curve, across both the primary and backup protection zones.

Asymmetry: the fault current is not purely symmetrical

The withstand equation is written in terms of RMS *symmetrical* current. A real fault, especially close to a source with high X/R ratio, carries a decaying DC offset in the first cycles that raises the actual $\int i^2 dt$ above the symmetrical value. The cleanest practice is to keep the withstand on its symmetrical basis and inflate the *fault* side by an asymmetry factor appropriate to the local X/R and the clearing time.

At longer clearing times (half a second and beyond) most of the DC has decayed and the effect is minor. At a few cycles with high X/R it is material and should not be neglected.

The piece most often missed: shields and neutrals

On shielded and concentric-neutral cable, the phase conductor is not the only metal that must survive a fault. The **metallic shield or concentric neutral** carries ground-fault return current, and it has its own short-circuit withstand — usually governed by a *separate* standard (ICEA P-45-482 for metallic shields and sheaths; ICEA S-94-649 for concentric-neutral construction) and a *lower* temperature ceiling, because a bare shield or neutral sits against the jacket and would damage it before reaching 250 °C.

The result is that such a cable has **two independent short-circuit limits**:

- the **phase conductor**, checked against the **phase fault** (three-phase or phase-to-phase);
- the **shield or neutral**, checked against the **ground-fault current that returns through it** (single-line-to-ground).

These answer to different fault currents, so the correct procedure is not "take the lower number." It is to verify each limit against its own duty. The shield/neutral check is frequently the binding one on solidly grounded systems, where ground-fault current is large; it is comfortable on impedance- or resistance-grounded systems, where it is small. And only the portion of ground-fault current that actually returns on a given shield or neutral counts — the rest divides among parallel neutrals, ground rods, and earth, a current-split problem shared with substation grounding analysis.

This dual-limit behavior deserves a full treatment of its own; the essential point here is simply that **the phase conductor rating does not cover the cable**. On shielded or concentric-neutral cable, check the metallic return path too.

The standards landscape

Standard	Scope
ICEA P-32-382	Short-circuit withstand of insulated <i>conductors</i> (the equations above)
ICEA P-45-482	Short-circuit performance of metallic <i>shields and sheaths</i>
ICEA S-94-649	Concentric-neutral cable construction, 5–46 kV
IEEE 242 (Buff Book)	Protection and coordination; cable damage curves in practice
IEC 60949	International equivalent, with an explicit heat-loss (non-adiabatic) correction

North American utility and industrial practice is built around the ICEA/IEEE family; international and IEC-specified projects use IEC 60949. They share the same adiabatic core, so results are close, but the conventions and the available correction terms differ. The governing standard should be taken from the project specification, not assumed.

When the adiabatic assumption breaks down

The adiabatic model is accurate and slightly conservative for fast faults — a few cycles up to roughly one to two seconds. Beyond that, real heat loss into the insulation becomes significant, and the adiabatic equation increasingly *underestimates* the cable's true capability, penalizing the design.

For longer durations, **IEC 60949** adds a correction factor (often written with an ϵ term) that accounts for the heat that escapes during the fault, yielding a higher and more realistic withstand. If a long clearing time is forcing an uneconomically large conductor, the non-adiabatic correction is the tool to claw back margin — provided the project's governing standard permits it.

A design checklist

1. **Identify the metal and insulation** → select K , λ , T_1 , T_2 (or the published F_C).
2. **Get the available fault current** from a system short-circuit study (symmetrical RMS).

3. **Apply asymmetry** for the local X/R and clearing time, on the fault side.
 4. **Get the actual total clearing time** — relay plus breaker, or fuse total-clear — not the tabulated 30 cycles.
 5. **Compute or look up the withstand** at that clearing time.
 6. **Compare:** withstand $\geq$ asymmetric available fault. Document the margin.
 7. **On shielded/concentric-neutral cable, repeat for the shield/neutral** against the ground-fault return current at the ground-fault clearing time.
 8. **For long clearing times**, consider the IEC 60949 non-adiabatic correction if the standard allows.
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Conclusion

Cable short-circuit withstand reduces to one compact adiabatic equation, but applying it well is a matter of supplying the right inputs: the correct temperatures from the insulation, the correct fault current from a system study, the correct *total* clearing time from the protection, and — on shielded cable — a second check on the metallic return path against the ground fault. The equation is simple; the discipline is in not letting a single tabulated number stand in for a careful, two-sided comparison.

All currents are RMS symmetrical unless stated. Withstand values follow the adiabatic ICEA P-32-382 form; short-circuit factors are representative values for the stated metal/insulation/temperature combinations and should be confirmed against the specific product and governing standard. The adiabatic model is conservative for fast faults and grows more conservative with duration, where IEC 60949's correction applies.